Hardware Preparation

The following document is a guide to modifying, flashing & operating the hardware shown below. The following programs will have to be installed on the PC before proceeding ahead.

- Arduino IDE:
  - https://www.arduino.cc/en/software/

The following information must also be readily available:

- Wi-Fi Network Name
- Wi-Fi Password Being Used

```
#define WIFI_CHANNEL     6

const char* ssid     = "Enter Network Name Here";
const char* password = "Enter Network Password Here";
const char* webhook  = "https://hoteldonotdisturb.tech/api/button/";

uint8_t receiverMAC[] = {0x80, 0xB5, 0x4E, 0xE8, 0x13, 0x50};
```

Fig 1. Wi-Fi Parameters

Before powering on make sure the files “Button_Sketch” & “Screen_Select_Sketch” are downloaded, unzipped and accessible. Open both files that ends in extension “.ino”. Once opened visit the section under “#define WiFi_Channel” and modify the “ssid” & “password fields accordingly for both files.” Save both files.

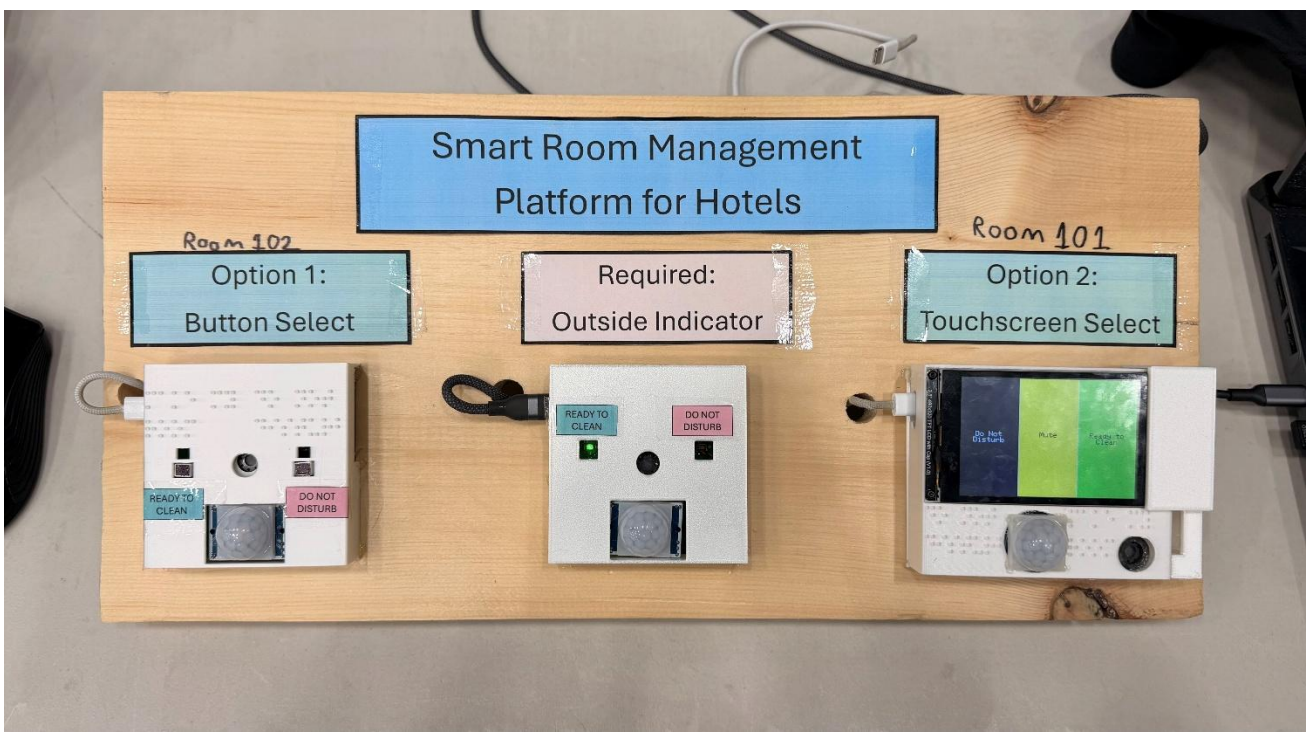

Fig 2. Hardware Layout

Next plug in either Option 1 or Option 2 into the PC. Determine which COMM port is being used by monitoring device manager. Un plug and re-plug to see which comm port disappears and reappears. This COMM number is different for every PC.

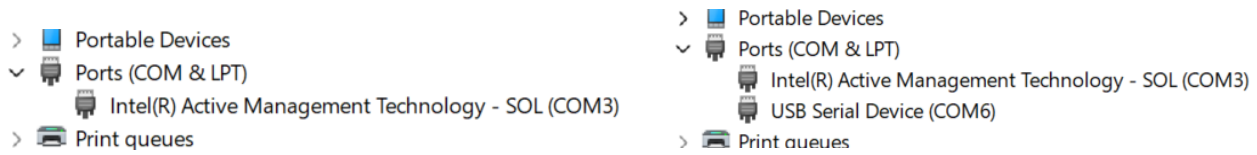

Fig 3. Device Manager Monitoring

Given that on this PC the hardware shows up as COM6. Note this number as it will be relevant on the next stage in Arduino IDE.

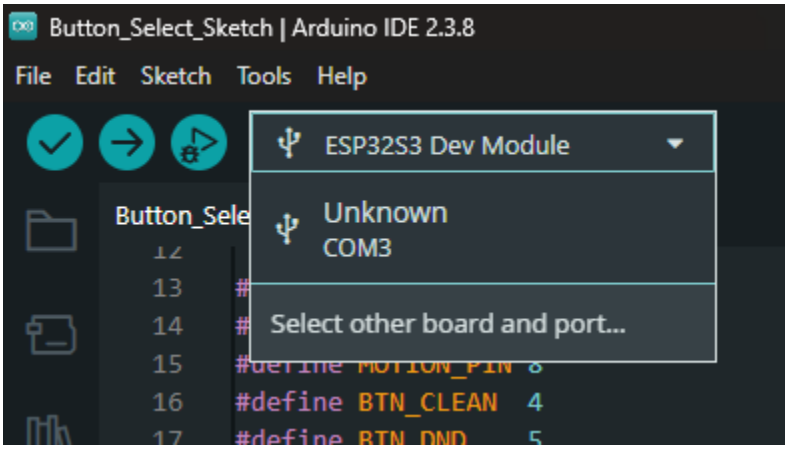

Fig 4. Flashing Method

In the figure above select the bottom option of “Select other board and port...”

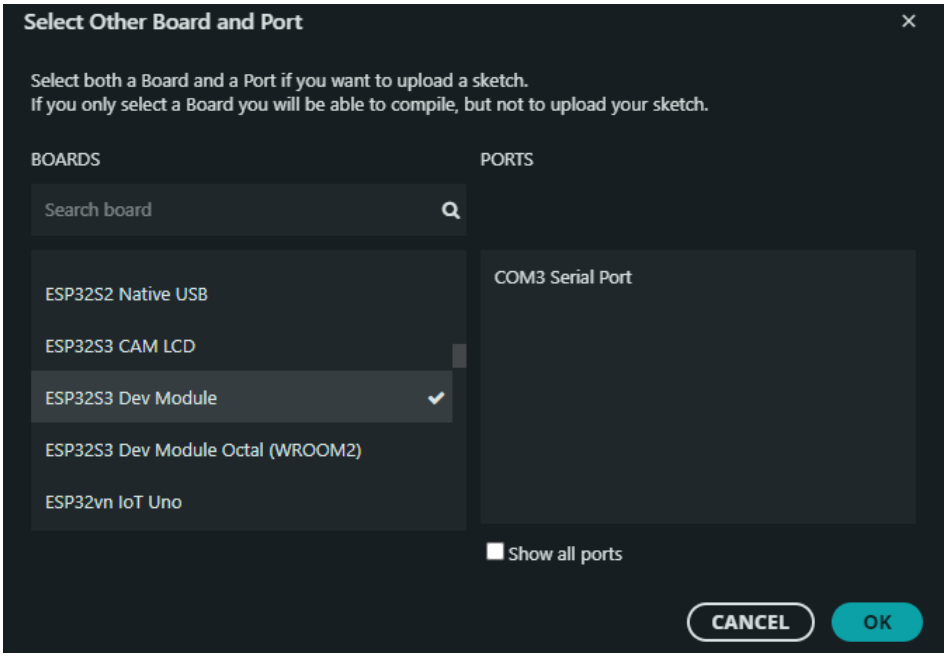

Fig 5. Establishing connection

Next as shown in the image above scroll down in the left panel to “ESP32S3 Dev Module” and select. On the right panel select the COMM port previously determined in Figure 3 and select OK in the bottom right corner.

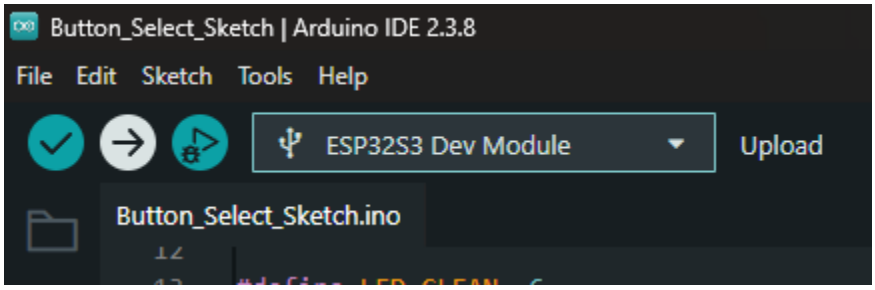

Fig 6. Uploading Code

Once connection has been established, select the upload button which is the arrow key pointing to the right between the check mark and build button. Repeat the steps above for the other option and make sure the appropriate COM port is chosen.

Now power on both the required module in the middle, and one of the two hardware options. Depending on whichever option selected it will take around 20 seconds to initialize and connect to Wi-Fi. The touch screen will show you the home screen where the user selects their option once connected to Wi-Fi. The button model will have both LEDs flashing and go dark once Wi-Fi has been initialized. The white dome is the motion sensor, and the circular cutout on all boards is the speaker outlet.